DETAILED PROJECT REPORT

for setting up a

CUSTOM HIRING CENTRE (CHC) OF FARM MACHINERY

Credit Linked Subsidy Scheme for Rural Entrepreneurs – individual applicant, TFO Rs 29.57 lakh

under the FARM MECHANIZATION SCHEME 2026-27

Department of Agriculture, Government of West Bengal



SUBMITTED BY

RAMPRASAD BISWAS

Individual applicant – farmer, Scheduled Caste (SC)

S/o Sudhir Biswas | Age: 60 years | Male

Vill. & P.O. Dakshin Durgapur, G.P. Deulbari Debipur, Block & P.S. Kultali, Dist. South 24 Parganas, PIN 743349

Mobile: 8116680624 | e-mail: surojit3136@gmail.com

Project site: own bastu land – Mouza Dakshin Durgapur, J.L. No. 37, Dag No. 317, G.P. Deulbari Debipur

PART A – MAIN REPORT

2. APPLICANT PROFILE

2.1 Details of the applicant

Particulars	Details
Name of the applicant	Ramprasad Biswas
Father's name	Sudhir Biswas
Applicant category	Individual – farmer (rural entrepreneur)
Age / gender	60 years / Male
Social category	Scheduled Caste (SC)
Address	Vill. Dakshin Durgapur, P.O. Dakshin Durgapur, G.P. Deulbari Debipur, Block & P.S. Kultali, Dist. South 24 Parganas, West Bengal, PIN 743349
Mobile / e-mail	8116680624 / surojit3136@gmail.com
Aadhaar No.	XXXX XXXX 7895 (masked)
Voter ID (EPIC) No.	WB/XX/XXX/XXX448 (masked)
PAN	XXXXXX147J (masked)
Bank account	State Bank of India, Kaikhali Branch – A/c No. XXXXXXX5724 (masked), IFSC SBIN0008785
Krishak Bandhu (Natun) / PM-KISAN registration No.	To be furnished – Krishak Bandhu (Natun) registration is required; PM-KISAN if any (Cl. XI(d)(i))
Occupation	Farmer; his wife is also a farmer. Neither of them is or was a Govt./semi-Govt. employee, a civic volunteer or contractual staff

2.2 Eligibility under FMS 2026-27 (CHC – individual applicant, Cl. XI(d) and XI(g))

Condition	Position of the applicant
Rural entrepreneur (farmer) registered under Krishak Bandhu (Natun); PM-KISAN registration if any (Cl. XI(d)(i))	Farmer of Vill. Dakshin Durgapur, G.P. Deulbari Debipur; Krishak Bandhu (Natun) registration No. to be furnished (Section 2.1)
TFO from Rs 20.00 lakh up to the SMAM maximum; subsidy for an individual Rs 16.00 lakh or 40% of TFO, whichever is less (Cl. XI(d)(i), XI(e)(ii))	TFO Rs 29.57 lakh; subsidy Rs 11,15,636 – below Rs 11,82,980 (40% of TFO)
Land in possession (own/leased/hired) of "commercial, bastu or viti" category; agricultural land not allowed (Cl. XI(d)(ii))	Own bastu land, 2 decimal (Mouza Dakshin Durgapur, J.L. No. 37, Khatian No. 682, Dag No. 317)
Applicant aged 18–70 years at the time of application (Cl. XI(d)(iv))	60 years
Applicant (and spouse) not an employee or retired employee of Govt./statutory body/Govt. undertaking/Panchayat/municipality/local body/Govt.-aided institution, nor a civic volunteer or contractual staff (Cl. XI(d)(iii); Annexure-V, item iv)	Yes – the applicant and his wife are farmers; neither of them holds or held any Govt. job, nor is a civic volunteer or contractual staff
Applicant and spouse have not availed any benefit under FSSM/OTA-SFI/FMB/FMH/CHC during 2022-23 to 2025-26 (Cl. XI(d)(v))	Yes – no benefit availed under any of these schemes during 2022-23 to 2025-26
At least four types of machinery/implements from Annexure-IV; at most two tractors (Cl. XI(g)(i), (iv))	4 types (one each); no tractor
Reservation: at least 22% of CHC benefits for Scheduled Caste and 7% for Scheduled Tribe applicants (Cl. XI(m)(iii))	Applicant is Scheduled Caste (SC)

3. PROJECT AT A GLANCE

3.1 Summary of the project

Item	Details
Applicant	Sri Ramprasad Biswas, S/o Sudhir Biswas – individual farmer, 60 years, Scheduled Caste
Scheme / component	Credit Linked Subsidy Scheme for Rural Entrepreneurs to set up Custom Hiring Centres (CHC) of Farm Machinery – Farm Mechanization Scheme 2026-27 (Memo No. 780-AG-11030(11)/1/2024-NAB-SEC dated 05.08.2026); individual applicant; credit-linked back-ended capital subsidy; approval by the DLSC (TFO up to Rs 60 lakh)
Project site	Vill. Dakshin Durgapur, P.O. Dakshin Durgapur, G.P. Deulbari Debipur, Block & P.S. Kultali, Dist. South 24 Parganas, PIN 743349
Land	Mouza Dakshin Durgapur, J.L. No. 37, Khatian No. 682, Dag No. 317, 2 decimal, Bastu; own land of the applicant
Shed	Not ready – a new open machinery shed (mandatory, Cl. XI(e)(i)) will be built at the project site; latest estimate Rs 1,10,035, included in TFO
Machinery (4 types)	Combine harvester (self-propelled, track type), power weeder, brush cutter, hand seeding machine – one each
Total Financial Outlay (TFO)	Rs 29,57,450 (Rs 29.57 lakh) = machinery Rs 27,89,515 + shed Rs 1,10,035 + AMC Rs 41,900 + insurance (first year) Rs 16,000
Admissible Govt. subsidy (back-ended)	Rs 11,15,636 = 37.72% of TFO (40% of the cost of each machine, the shed, the AMC and the insurance, subject to the machine-wise ceilings of Annexure-IV; the combine harvester is held to its ceiling of Rs 10,00,000) – within the individual limit of Rs 16.00 lakh or 40% of TFO, whichever is less (Cl. XI(e)(ii)); released in two halves to the Subsidy Reserve Fund (Cl. XI(k))
Applicant's margin money	Rs 7,39,362 (25% of TFO – the minimum under Cl. XI(e)(v)), deposited with the bank before the subsidy is released
Bank term loan	Rs 11,02,452 with interest (the bank loan in the application) – repaid by the applicant; the bank also lends Rs 11,15,636 without interest against the subsidy (kept in the Subsidy Reserve Fund), so the total loan sanctioned is Rs 22,18,088 = TFO less margin (Annexure-IX; Section 9.3)
Financing bank	State Bank of India, Kaikhali Branch (IFSC SBIN0008785) – a commercial bank (Cl. XI(b)); interest assumed at 9% p.a. (the AIF lending-rate cap), 6% p.a. net after the 3% AIF interest subvention
AIF convergence	Yes – AIF benefit to be availed on the loan with interest: 3% p.a. interest subvention (Section 9.4)
Service area / crop	Deulbari Debipur G.P. first, then the other G.P.s of Kultali block listed in Section 6.1 – main crop paddy (Aman and Boro)
Implementation	CHC to be set up within 3 months of sanction or by 31 March 2027, whichever is earlier (Cl. XI(k)(xiii)); first full year of operation 2027-28 (Year I)
Area and bookings	About 158 ha in Year I rising to about 211 ha a year from Year IV; about 634 to 845 hire bookings a year
Employment	Combine operator and helper (seasonal, about 52 to 69 paid days each a year); the applicant runs the CHC himself

4. PROJECT OBJECTIVE & LOCAL NEED

4.1 Background and local need

Farm mechanization raises production and productivity, makes farm operations timely and reduces the cost of cultivation. In Kultali block of South 24 Parganas, in the Sundarbans area, paddy is the main crop and most farmers are small and marginal. They cannot buy costly machines such as a combine harvester, and at harvest time labour is scarce and expensive. As a result, harvesting is delayed and the crop is exposed to rain, cyclones and shattering losses.

To address this, Sri Ramprasad Biswas, a farmer of Vill. Dakshin Durgapur, proposes to establish a Custom Hiring Centre (CHC) of farm machinery on his own bastu land in the village, to hire out machines to local farmers at reasonable charges with preference to small and marginal farmers, as intended under Cl. XI(c) of the FMS 2026-27 guideline.

Paddy is grown in two seasons – Aman (kharif) and Boro (summer) – so the combine harvester has two harvest windows a year; pulses, oilseeds and vegetables give work for the small implements (Section 7.2).

4.2 Objectives of the project

Primary objectives

- To give a combine harvester and small farm implements on hire to farmers at village level, with priority to small and marginal farmers.
- To complete paddy harvesting on time and reduce crop losses.
- To reduce the cost of harvesting and the dependence on scarce labour.
- To create regular self-employment and income for the applicant, and seasonal jobs for local youth as operator and helper.
- To increase farm-power availability in Kultali block.

Secondary objectives

- To spread awareness among farmers about modern farm machinery.
- To help farmers get information on Government schemes for agriculture.

5. PROJECT SITE, LAND & SHED DETAILS

5.1 Project site and land

Particulars	Details
Location of the CHC	Vill. Dakshin Durgapur, P.O. Dakshin Durgapur, G.P. Deulbari Debipur, Block & P.S. Kultali, Dist. South 24 Parganas, PIN 743349
Mouza / J.L. No.	Dakshin Durgapur / 37
Khatian No.	682
Plot (Dag) No.	317
Area	2 decimal
Classification of land	Bastu – eligible category under Cl. XI(d)(ii)
Possession	Own land of the applicant
Approach road	8 ft wide

5.2 Machinery shed

Particulars	Details
Status	Not ready – a new shed will be constructed at the project site
Ownership / location	Own open shed – mandatory for a CHC (Cl. XI(e)(i)) – to be built by the applicant on his own project land
Type and size	Length: 30 ft Width: 20 ft Height: 12 ft - Covered area: 600 sq. ft.
Cost included in TFO	Rs 1,10,035 (latest estimate) = 3.72% of TFO – within the 5% limit for shed cost (Rs 1,47,873) and the 10% limit for ancillary cost (Cl. XI(e)(iv)); vetted plan and estimate enclosed (Cl. XI(f), item 10)
Subsidy on the shed	Rs 44,014 (40% of cost)
Construction	To be built and completed before the machinery is delivered and jointly inspected

The project site is the applicant's own bastu land in his home village, Dakshin Durgapur (G.P. Deulbari Debipur).

6. SERVICE AREA, MACHINERY DEMAND & BENEFICIARY FARMERS

6.1 Service area

The CHC is in Deulbari Debipur G.P. of Kultali block (BDO, Kultali). It will serve the farmers of Deulbari Debipur G.P. first, with preference to small and marginal farmers, and then farmers of the other areas listed in the application: Gopalganj, Gurguria Bhubaneswari, Jalaberia-I, Jalaberia-II, Maipith Baikunthapur, Merigunj-I, Merigunj-II and Kundakhali Godabar G.P.s and Kultali Municipality.

Particulars	Details
Primary service area	Deulbari Debipur G.P. – Vill. Dakshin Durgapur (CHC site and the applicant's village), Deulbari Debipur, Katamari, Madhabpur and Shyamnagar
Secondary service area	The other G.P.s and areas of Kultali block listed above, after local demand is met; machines may also go to nearby blocks, but not outside West Bengal (Cl. XI(k)(xv))
Main crop	Paddy – Aman (kharif) and Boro (summer)

6.2 Planned harvesting service (combine harvester)

Year	Capacity utilisation	Working hours	Working days (8 h/day)	Area harvested (ha) at 0.44 ha/hour
I (2027-28)	60%	360	45	158.4
II (2028-29)	70%	420	53	184.8
III (2029-30)	75%	450	56	198.0
IV–VII (2030-31 to 2033-34)	80%	480	60	211.2

Estimated field capacity of the combine harvester: 0.40 to 0.48 ha/hour. Area = hours × 0.44 ha/hour (average); at 0.40–0.48 ha/hour the area is 144–173 ha in Year I and 192–230 ha from Year IV. Bookings = area ÷ 0.25 ha; 75 days = Aman 50 + Boro 25.

6.3 Beneficiary farmers

Particulars	Year I	Year IV onward
Hire bookings – combine harvester (no.)	634	845
Area harvested (ha)	158	211
Farm families served (approx., indicative)	about 330	about 440
Farmers using the small implements (no.)	about 80	about 110

6.4 Proposed hire charges (Year I)

Machine	Hire charge	Equivalent
Combine harvester	Rs 3,600 per hour	About Rs 8,180 per ha (about Rs 1,100 per bigha) at 0.44 ha/hour (Rs 7,500–9,000 per ha at 0.48–0.40 ha/hour)
Power weeder	Rs 250 per hour	–
Brush cutter	Rs 150 per hour	–
Hand seeding machine	Rs 100 per day	–

Rates rise 3% a year; actual "reasonable charges" are fixed in consultation with the ADA (Block) and the Asstt. DA (Admn) of the sub-division (Annexure-V, item xiv) and shown on the signboard with contact details (Annexure-V, item xi).

7. PROPOSED MACHINERY & TECHNICAL JUSTIFICATION

7.1 Machinery proposed

Sl	Machine and Annexure-IV category	Make, model and dealer	Specification	Qty	Indicative price / cost (Rs)
1	Combine Harvester – self-propelled, track type (Annexure-IV item 19(g): Combine Harvester (Track type) > 6 feet cutter bar width)	Escorts Kubota Ltd. Kubota DC-68G Harvesking Pro Dealer: Bengal Agro Machinery Corporation, Purba Bardhaman	Self-propelled paddy combine harvester (track type); estimated field capacity 0.40–0.48 ha/hour	1	26,68,356
2	Power Weeder (Annexure-IV item 6(c): Power Weeder, engine operated 5 BHP & below 7.5 BHP)	Ralli Agrotech Ltd. Ralli RRT-100P Dealer: Bharat Tractors, Nadia	5.3 hp, 212 cc, petrol engine driven	1	89,397
3	Brush Cutter (Annexure-IV item 46: Brush Cutter (electric/engine powered))	Ralli Agrotech Ltd. Ralli BC-52 Dealer: Bharat Tractors, Nadia	52 cc, single cylinder, 2-stroke petrol engine	1	21,367
4	Hand Seeding Machine (Annexure-IV item 35: Hand Seedling Machine)	Ralli Agrotech Ltd. Ralli RS-12 Dealer: Bharat Tractors, Nadia	Manually operated	1	10,395
	Total cost of machinery (4 types)			**4**	**27,89,515**
5	Add: shed (Section 5.2)			1	1,10,035
6	Add: AMC – paid upfront to the dealer, 18 months (Section 8.3)			–	41,900
7	Add: insurance of the machinery – first year (Section 8.3)			–	16,000
	Total Financial Outlay (TFO) = machinery + shed + AMC + insurance				**29,57,450**

7.2 Technical justification

Machine	Operation covered	Why it is needed and how it is sized
Combine harvester (Kubota DC-68G Harvesking Pro)	Reaping, threshing and cleaning of paddy in one pass (Aman and Boro)	Removes the harvest-labour bottleneck, so mature paddy is less exposed to rain, cyclones and shattering. Track type for soft, wet fields. One unit matches the 75-day window (600 hours at 8 h/day).
Power weeder (Ralli RRT-100P)	Weeding / inter-cultivation and light soil working in small plots	Cuts manual weeding labour and cost; 5.3 hp suits small plots; 150 hours a year at full use. Scope: vegetables and other upland crops in the service area.
Brush cutter (Ralli BC-52)	Cutting weeds and grass on bunds, field boundaries and channels; cutting fodder and crop residue	Low-cost and portable; keeps bunds and channels clean before sowing or transplanting; 150 hours a year at full use.
Hand seeding machine (Ralli RS-12)	Line sowing of seed at uniform spacing and depth	Saves seed and sowing labour over broadcasting; manual, no fuel; 50 days a year at full use. Scope: line sowing of pulses, oilseeds and vegetables in the service area.

Together the four machines cover sowing (hand seeding machine), weeding and light tillage (power weeder), harvesting and threshing (combine harvester) and cutting of weeds, fodder and crop residue (brush cutter) – the end-to-end use of farm machinery sought for a CHC (Cl. XI(c)(i), XI(g)(iv)).

8. OPERATING PLAN & EMPLOYMENT

8.1 Operating calendar (indicative)

Period	Combine harvester	Small implements
April - May	Boro paddy harvest (local and nearby blocks within West Bengal)	Brush cutter for small plots
June - October	Servicing under AMC; standby	Power weeder and hand seeder in kharif vegetables and line-sown crops
November - December	Aman paddy harvest - peak season	Brush cutter for small plots and bunds
January - March	Off-season overhaul under AMC	Power weeder and hand seeder in rabi vegetables

8.2 Staffing and employment

Position	No.	Engagement	Wage (Year I)	Paid days a year	Annual cost, Year I (Rs)
Combine operator (skilled)	1	Hired, seasonal	Rs 2,300/day	52 (Year I) to 69 (Year IV on)	1,19,600
Helper	1	Hired, seasonal	Rs 600/day	52 (Year I) to 69 (Year IV on)	31,200
Total paid staff	**2**				**1,50,800**

Paid days = working days + 15% for travel and standby, rounded to whole days; wages rise 5% a year. The applicant manages the CHC himself and draws no salary; the small implements are operated by him or by local workers paid only for that work, within their running cost.

8.3 Maintenance and insurance

- Annual Maintenance Contract (AMC) with the dealer: Rs 73,900 for the first 30 months (Rs 29,560 a year in Years I–II and Rs 14,780 for the first 6 months of Year III); renewal at Rs 40,000 a year from month 31 (Rs 20,000 in Year III), rising 5% a year from Year V. Rs 41,900 of the AMC is included in the TFO and paid upfront to the dealer (it covers the Year I charge of Rs 29,560 and Rs 12,340 of Year II); the balance of Rs 32,000 and the renewals are paid from hire income.
- Repairs and spares (one row, apart from the AMC): worked out by the Indian Standard IS 9164:1979 (BIS) from the hours each machine works, so the cost rises as the machines get older. The AMC is part of this maintenance cost, so it is deducted and not counted twice (details under FA-2).
- Insurance of the machinery: Rs 16,000 for one year; renewed every year and taken at Rs 16,000 a year. The first year's premium (Rs 16,000) is included in the TFO and paid upfront; the renewals from Year II are paid from hire income.
- Pre-season servicing before each harvest; the machines are parked under the shed when not in use.

8.4 Display, records and use of the machines

- A permanent signboard at the CHC (name of the CHC and its proprietor, year of financing, subsidy, financing bank and credit received, hire rates and contact details), a metal plate on each machine, a money receipt or cash/credit memo to every user and a hire register (Cl. XI(k)(xvi)–(xviii); Annexure-V, items vii, xi and xiii).
- Machines are hired out to all farmers, small and marginal farmers first; they may go to nearby blocks or districts after local demand is met, but never outside West Bengal (Cl. XI(k)(xv)). The CHC will not be sold, leased or mortgaged within 6 years (Cl. XI(j)(iii), (vii)).

9. PROJECT COST, SUBSIDY & MEANS OF FINANCE

9.1 Total Financial Outlay (TFO)

Component (as in Cl. XI(e)(i))	Included in TFO (Rs)	Remarks
Indicative cost of machinery (4 types – Section 7)	27,89,515	
Cost of shed (new open machinery shed, to be built – Section 5.2)	1,10,035	3.72% of TFO – within the 5% limit for shed cost (Rs 1,47,873) and the 10% limit for ancillary cost (Cl. XI(e)(iv)); open shed is mandatory (Cl. XI(e)(i)); vetted plan and estimate enclosed (Cl. XI(f), item 10)
AMC	41,900	Included; paid upfront to the dealer – part of the AMC of Rs 73,900 for 30 months (covers the Year I charge of Rs 29,560 and Rs 12,340 of Year II); the balance of Rs 32,000 and the renewals are paid from hire income
Insurance	16,000	Included; comprehensive insurance of the machinery for the first year (Year I); renewals from Year II (Rs 16,000 a year) are paid from hire income
Total Financial Outlay (TFO)	**29,57,450**	**Rs 29.57 lakh – within the CHC range of Rs 20.00 lakh to the SMAM maximum (Cl. XI(e)(ii)); assistance @40% of TFO subject to machine-wise ceilings (Cl. XI(e)(i)); shed + AMC + insurance Rs 1,67,935 = 5.68% of TFO, within the 10% limit for ancillary cost (Cl. XI(e)(iv))**

9.2 Admissible subsidy (Annexure-IV)

Sl	Item	Price / cost (Rs)	40% of cost (Rs)	Ceiling – Annexure-IV (Rs)	Admissible subsidy (Rs)
1	Combine Harvester	26,68,356	10,67,342	10,00,000	10,00,000
2	Power Weeder	89,397	35,758	60,000	35,758
3	Brush Cutter	21,367	8,546	20,000	8,546
4	Hand Seeding Machine	10,395	4,158	10,000	4,158
5	New open machinery shed	1,10,035	44,014	–	44,014
6	AMC (paid upfront)	41,900	16,760	–	16,760
7	Insurance (first year)	16,000	6,400	–	6,400
	Total	**29,57,450**	**–**	**–**	**11,15,636**

Each machine: 40% of cost or its Annexure-IV ceiling, whichever is less (rounded down to the rupee, as on the portal); the combine harvester is held to its ceiling of Rs 10,00,000. Total Rs 11,15,636 = 37.72% of TFO – within the individual limit of Rs 16.00 lakh or 40% of TFO (Rs 11,82,980), whichever is less, and above the minimum subsidy of Rs 8.00 lakh (Cl. V(3), XI(d)(i), XI(e)(ii)).

9.3 Means of finance

Sl	Source	Amount (Rs)	% of TFO
1	Applicant's margin money (minimum 25% of TFO, Cl. XI(e)(v))	7,39,362	25.00%
2	Total bank term loan sanctioned (TFO less margin, Cl. XIII(3)) – State Bank of India, Kaikhali Branch, under AIF (Section 9.4)	22,18,088	75.00%
	(a) loan without interest (Annexure-IX, item 7) – backed by the Govt. subsidy kept in the Subsidy Reserve Fund (SRF), locked in for 5 years	11,15,636	37.72%
	(b) loan with interest (Annexure-IX, item 8) – the bank loan in the application, repaid by the applicant (9% p.a. less 3% AIF subvention = 6% p.a. net)	11,02,452	37.28%
	Total = TFO	**29,57,450**	**100.00%**

The subsidy goes to the Subsidy Reserve Fund in two halves – 50% (Rs 5,57,818) as advance after the loan is sanctioned and the margin money deposited, and 50% (Rs 5,57,818) after joint inspection and geo-tagging of the machines (Cl. XI(k)(vi), (x), (xi)). It stays locked in for 5 years and is adjusted after the loan with interest is repaid; interest is charged only on the loan with interest (Cl. XI(j)(ii), (iv); XIII(3), (5)).

9.4 Convergence with the Agriculture Infrastructure Fund (AIF)

- A Custom Hiring Centre with at least 4 farm machines/implements is an eligible project under AIF, and the FMS feasibility report records AIF convergence (Annexure-XIV, item 10); this CHC has 4 machines, including a combine harvester.
- AIF gives an interest subvention of 3% p.a. on loans up to Rs 2 crore for a maximum of 7 years, and credit-guarantee cover through CGTMSE for eligible borrowers.
- Repayment under AIF can be up to 7 years including a moratorium of up to 2 years; this project follows the FMS rule of one grace year and repayment within 5 years (Cl. XI(j)(ii); XIII(6)(e)) – 10 half-yearly instalments in Years II–VI (FA-4).

10. EXPECTED ANNUAL INCOME, EXPENDITURE & NET RETURN

10.1 Total Financial Outlay and expected annual return (Annexure-VIII, items 16 and 17)

Sl	Item	Amount (Rs)	Basis / where to verify
1	Total Financial Outlay (TFO)	29,57,450	Machinery Rs 27,89,515 + new open shed Rs 1,10,035 + AMC Rs 41,900 + first-year insurance Rs 16,000 (Section 9.1)
2	(a) Gross annual income (Year I)	13,35,000	Hire income in 2027-28 at 60% utilisation: combine Rs 12,96,000 + small implements Rs 39,000 (Section 10.2)
3	(b) Total expenditure (Year I)	6,92,224	Operating costs: fuel and lubricants, operator and helper, shifting, small implements, repairs and spares, AMC, insurance, miscellaneous (Section 10.4); AMC and insurance at full cost, of which Rs 45,560 is already paid from the TFO (Section 10.3)
4	**(c) Net income (Year I) = (a) – (b)**	**6,42,776**	**Before interest Rs 66,147 (net of the 3% AIF subvention), depreciation Rs 4,29,431 and income tax (nil) (Section 10.3)**

10.2 Hire income in Year I (2027-28, 60% utilisation)

Machine	Basis	Income (Rs)
Combine harvester	360 hours × Rs 3,600	12,96,000
Power weeder	90 hours × Rs 250	22,500
Brush cutter	90 hours × Rs 150	13,500
Hand seeding machine	30 days × Rs 100	3,000
Total		**13,35,000**

10.3 Annual income, expenditure and net return

	Particulars	Year I (2027-28)	Year IV (2030-31, 80%)
A	Gross hire income	13,35,000	19,45,142
B	Operating expenditure (fuel and lubricants, operator and helper, shifting, small-implement running, repairs and spares, AMC, insurance, miscellaneous)	6,92,224	11,48,359
C	**Net operating income (A – B)**	**6,42,776**	**7,96,783**
D	Interest on bank loan (6% p.a. net, after 3% AIF subvention)	66,147	36,381
E	Depreciation (non-cash)	4,29,431	2,64,989
F	Net profit before tax	1,47,198	4,95,413
G	Income tax (projected – nil after the rebate, FA-3)	0	0
H	**Net profit after tax**	**1,47,198**	**4,95,413**
I	Add: AMC and insurance paid upfront from TFO (included in B; no cash outflow in the year)	45,560	0
J	Cash surplus (H + E + I)	6,22,189	7,60,402
K	Loan principal repaid	0 (grace year)	2,20,490
L	**Net cash after loan repayment**	**6,22,189**	**5,39,912**

For the application form (Annexure-VIII): item 13 – area of operation about 158 ha in Year I, rising to about 211 ha a year from Year IV; item 14 – combine harvester, power weeder, brush cutter and hand seeding machine; item 16 – TFO Rs 29,57,450; item 17 (Year I) – (a) gross annual income Rs 13,35,000, (b) total expenditure Rs 6,92,224, (c) net income Rs 6,42,776.

10.4 Operating expenditure, Year I (Rs)

Fuel and lubricants	Operator and helper	Shifting of combine	Small implements running	Repairs and spares	AMC	Insurance	Miscellaneous	Total
3,09,600	1,50,800	45,424	28,545	54,295	29,560	16,000	58,000	6,92,224

11. SWOT / RISK MANAGEMENT

11.1 SWOT analysis

	Points
Strengths	1. The applicant is a farmer of the same village and knows the local farmers and their needs. 2. Paddy is the main crop, so combine harvesting is in demand in both Aman and Boro seasons. 3. The admissible Govt. subsidy (37.72% of TFO) and the 3% AIF interest subvention reduce the debt burden; average DSCR 3.37 (FA-5). 4. Own bastu land at the project site for the CHC and its new shed.
Weaknesses	1. Depends on a hired operator for the combine harvester. 2. Most of the income comes from one machine (the combine harvester) in two short harvest seasons.
Opportunities	1. Shortage of farm labour at harvest time. 2. Two paddy seasons (Aman and Boro) give two harvest windows a year.
Threats	1. Rise in diesel price. 2. Machine breakdown during the harvest season. 3. Operator not available. 4. Rain or cyclone shortening the harvest window.

11.2 Risk management

Risk	Mitigation
Rise in diesel price	Annual review of hire charges with the ADA (Block); fuel-efficient operation and route planning by village cluster.
Breakdown during the harvest season	AMC with the dealer (Rs 73,900 for 30 months – Rs 41,900 of it paid upfront from TFO – then renewal); 1-year warranty and 4-year after-sales service under the guideline (Cl. VIII(B)(i)); pre-season servicing; stock of fast-moving spares.
Operator not available	Written agreement with an experienced operator before each season; a back-up operator trained from local youth.
New to running a CHC	Supplier demonstration and training at delivery; guidance from the ADA; simple records (hire register, log book, money receipts).
Accident, damage or theft	Comprehensive insurance (Rs 16,000 a year; first year paid from TFO); parking under the shed; geo-tracking through the telematics kit (Cl. VIII(B)(w)).

PART B – FINANCIAL ANNEXURE

FA-1 Capacity and hire income (Rs)

Particulars	I 2027-28	II 2028-29	III 2029-30	IV 2030-31	V 2031-32	VI 2032-33	VII 2033-34	Total
Capacity utilisation	60%	70%	75%	80%	80%	80%	80%	–
Harvester working hours	360	420	450	480	480	480	480	3,150
Hire charge per hour (Rs)	3,600	3,708	3,819	3,934	4,052	4,173	4,299	–
Harvester hire income	12,96,000	15,57,360	17,18,550	18,88,320	19,44,960	20,03,040	20,63,520	1,24,71,750
Small implements hire income	39,000	46,865	51,719	56,822	58,526	60,282	62,091	3,75,305
Total hire income	**13,35,000**	**16,04,225**	**17,70,269**	**19,45,142**	**20,03,486**	**20,63,322**	**21,25,611**	**1,28,47,055**

Basis – combine harvester	Details	Small implement	Use at 100% capacity	Hire charge	Income at 100% (Rs)
Number of units	1	Power weeder	150 hours	Rs 250 per hour	37,500
Hours of operation a day	8	Brush cutter	150 hours	Rs 150 per hour	22,500
Harvest-season window	75 days (Aman 50 + Boro 25)	Hand seeding machine	50 days	Rs 100 per day	5,000
Working hours at 100% capacity	600 hours a year	Total			65,000

Harvester income = working hours × hire charge per hour. Small-implement income = Rs 65,000 × utilisation × 1.03^(year − 1).

FA-2 Operating expenditure (Rs)

Particulars	I 2027-28	II 2028-29	III 2029-30	IV 2030-31	V 2031-32	VI 2032-33	VII 2033-34	Total
Fuel and lubricants	3,09,600	3,72,036	4,10,568	4,51,078	4,64,610	4,78,548	4,92,905	29,79,345
Operator and helper wages	1,50,800	1,82,700	2,07,821	2,31,641	2,43,223	2,55,384	2,68,153	15,39,722
Harvester shifting charges	45,424	52,635	60,238	65,147	67,102	69,115	71,188	4,30,849
Small implements running and fuel	28,545	34,617	38,557	42,758	44,458	46,229	48,075	2,83,239
Repairs and spares (as per IS 9164)	54,295	1,34,151	1,86,099	2,34,593	2,65,610	2,91,903	3,14,876	14,81,527
AMC (Rs 73,900 for 30 months, then renewal; Rs 41,900 paid from TFO)	29,560	29,560	34,780	40,000	42,000	44,100	46,305	2,66,305
Insurance (Rs 16,000 a year; Year I paid from TFO)	16,000	16,000	16,000	16,000	16,000	16,000	16,000	1,12,000
Miscellaneous expense (semi-fixed)	58,000	60,900	63,945	67,142	70,499	74,024	77,726	4,72,236
Total operating expenditure	**6,92,224**	**8,82,599**	**10,18,008**	**11,48,359**	**12,13,502**	**12,75,303**	**13,35,228**	**75,65,223**

Fuel – combine harvester	Details	Manpower (Year I)	No.	Wage	Paid days	Annual cost (Rs)
Fuel requirement per hour	8 litres	Combine harvester operator	1	Rs 2,300/day	52	1,19,600
Working hours at 100% capacity	600 hours	Helper	1	Rs 600/day	52	31,200
Fuel requirement at 100% capacity	4,800 litres a year	Total	2			1,50,800
Fuel cost at 100% capacity (Rs 100/litre)	Rs 4,80,000					
Year I fuel (60%) + lubricants 7.5%	Rs 2,88,000 + Rs 21,600 = Rs 3,09,600					

FA-3 Depreciation and income tax (Rs)

Book depreciation (WDV)	I 2027-28	II 2028-29	III 2029-30	IV 2030-31	V 2031-32	VI 2032-33	VII 2033-34	Total
Machinery @ 15%	4,18,427	3,55,663	3,02,314	2,56,967	2,18,422	1,85,658	1,57,810	18,95,261
Shed @ 10%	11,004	9,903	8,913	8,022	7,219	6,497	5,848	57,406
Total depreciation for the year	**4,29,431**	**3,65,566**	**3,11,227**	**2,64,989**	**2,25,641**	**1,92,155**	**1,63,658**	**19,52,667**
Closing value – machinery	23,71,088	20,15,425	17,13,111	14,56,144	12,37,722	10,52,064	8,94,254	–
Closing value – shed	99,031	89,128	80,215	72,193	64,974	58,477	52,629	–
Closing value – total	24,70,119	21,04,553	17,93,326	15,28,337	13,02,696	11,10,541	9,46,883	–

Income tax computation	I 2027-28	II 2028-29	III 2029-30	IV 2030-31	V 2031-32	VI 2032-33	VII 2033-34	Total
Net profit before tax	1,47,198	2,93,220	3,91,424	4,95,413	5,41,191	5,85,942	6,26,725	30,81,113
Add: depreciation as per books	4,29,431	3,65,566	3,11,227	2,64,989	2,25,641	1,92,155	1,63,658	19,52,667
Add: AMC and insurance paid from TFO – 40% met by subsidy, not allowed for tax	18,224	4,936	0	0	0	0	0	23,160
Less: tax depreciation – machinery (15% WDV on Rs 17,41,053)	2,61,158	2,21,984	1,88,687	1,60,384	1,36,326	1,15,877	98,496	11,82,912
Less: tax depreciation – shed (10% WDV on Rs 66,021)	6,602	5,942	5,348	4,813	4,332	3,898	3,509	34,444
Taxable profit (total income)	**3,27,093**	**4,35,796**	**5,08,616**	**5,95,205**	**6,26,174**	**6,58,322**	**6,88,378**	**38,39,584**
Tax at new-regime slab rates (s. 202)	0	1,790	5,431	9,760	11,309	12,916	14,419	55,625
Less: rebate u/s 156 (up to Rs 12 lakh)	0	1,790	5,431	9,760	11,309	12,916	14,419	55,625
Income tax payable (incl. 4% cess)	0	0	0	0	0	0	0	0

Opening value = cost in Year I, then the previous year-end value. Book depreciation is on the full cost of the machinery and the shed; the subsidy is shown as a capital grant in the balance sheet. For tax, depreciation is on cost net of subsidy, and the AMC and insurance paid from the TFO are allowed only for the 60% not met by the subsidy. Hire income of the CHC is business income, not agricultural income. For projection, the applicant is taken as a resident individual under the new tax regime (s. 202, Income-tax Act, 2025: nil up to Rs 4 lakh, 5% on Rs 4–8 lakh, 10% on Rs 8–12 lakh) with no other taxable income; as his total income stays below Rs 12 lakh in every year, the rebate under s. 156 (up to Rs 60,000) makes the tax nil. The actual tax will depend on his total income and the Income-tax law in force.

FA-4 Repayment of the loan with interest, with AIF interest subvention (Rs)

Year	Opening balance	Interest @ 9%	Less: AIF subvention @ 3%	Net interest @ 6% (paid by the applicant)	Principal	Closing balance
I (2027-28)	11,02,452	99,221	33,074	66,147	0 (grace)	11,02,452
II (2028-29)	11,02,452	94,260	31,420	62,840	2,20,490	8,81,962
III (2029-30)	8,81,962	74,416	24,806	49,610	2,20,490	6,61,472
IV (2030-31)	6,61,472	54,571	18,190	36,381	2,20,490	4,40,982
V (2031-32)	4,40,982	34,727	11,575	23,152	2,20,490	2,20,492
VI (2032-33)	2,20,492	14,883	4,961	9,922	2,20,492	0
Total		**3,72,078**	**1,24,026**	**2,48,052**	**11,02,452**	

Interest assumed at 9% p.a. (the AIF lending-rate cap; the actual rate is set by the bank, Cl. XIII(6)(c)) on the reducing balance. The loan with interest (Rs 11,02,452) is repaid in 10 half-yearly instalments in Years II–VI (Rs 1,10,245 each, the last Rs 1,10,247) – one grace year, then repayment over 5 years and not earlier (Cl. XI(j)(ii), (vi); Annexure-V, item x). The 3% AIF subvention is claimed by the bank and passed on to the applicant, so he bears 6% p.a. – the figure used in Section 10.3 and FA-3. No interest is charged on the loan without interest (Rs 11,15,636), which is adjusted from the Subsidy Reserve Fund after the loan with interest is repaid (Cl. XIII(3), (5)).

FA-5 Debt service coverage ratio (DSCR) (Rs)

Debt service coverage	I 2027-28	II 2028-29	III 2029-30	IV 2030-31	V 2031-32	VI 2032-33	Total (I–VI)
Net profit after tax	1,47,198	2,93,220	3,91,424	4,95,413	5,41,191	5,85,942	24,54,388
Add: depreciation (non-cash)	4,29,431	3,65,566	3,11,227	2,64,989	2,25,641	1,92,155	17,89,009
Add: net interest on the bank loan	66,147	62,840	49,610	36,381	23,152	9,922	2,48,052
Add: AMC and insurance paid from TFO (non-cash)	45,560	12,340	0	0	0	0	57,900
(A) Cash available for debt service	**6,88,336**	**7,33,966**	**7,52,261**	**7,96,783**	**7,89,984**	**7,88,019**	**45,49,349**
Net interest (6% p.a.)	66,147	62,840	49,610	36,381	23,152	9,922	2,48,052
Principal repaid	0	2,20,490	2,20,490	2,20,490	2,20,490	2,20,492	11,02,452
(B) Total debt service	**66,147**	**2,83,330**	**2,70,100**	**2,56,871**	**2,43,642**	**2,30,414**	**13,50,504**
DSCR = (A) ÷ (B)	**10.41**	**2.59**	**2.79**	**3.10**	**3.24**	**3.42**	**3.37 (average)**

DSCR = cash available for debt service (A) ÷ net interest and principal (B). Year I is the grace year (interest only). Average DSCR for Years I–VI = 3.37 (Years II–VI: 3.01) – comfortably above the 1.25–1.50 range that banks usually require; credit merit is a selection criterion for CHC (Cl. XI(i)(iv)).